Area and differential equation 6]

Q1. Find the continuous function which satisfies the relation $\int_0^x t f(x-t)\,dt = \int_0^x f(t)\,dt + \sin x + \cos x - x - 1$ for all real number x.

Q2. In a college hostel accommodating 1000 students, one of them came in carrying a flue virus, then the hostel was isolated. If the rate at which the virus spreads is assumed to be proportional not only to the number N_1 of infected students, but also to the number of non-infected students, and of the number of infected students is 50 after 4 days, then show that more than 95% of the students will be infected after 10 days.

Q3. C: Let $f:[0,1] \to R$ (the set of all real numbers) be a function. Suppose the function f is twice differentiable, $f(0) = f(1) = 1$ and satisfies $f''(x) - 2f'(x) + f(x) \geq e^x$, $x \in [0,1]$. Which of the following is true for $0 < x < 1$?

(a) $0 < f(x) < \infty$ (b) $-\frac{1}{2} < f(x) < \frac{1}{2}$

(c) $-\frac{1}{4} < f(x) < 1$ (d) $-\infty < f(x) < 0$

Q4. If the function $e^{-x} f(x)$ assumes its minimum in the interval $[0,1]$ at $x = \frac{1}{4}$, which of the following is true?

(a) $f'(x) < f(x)$, $\frac{1}{4} < x < \frac{3}{4}$ (c) $f'(x) < f(x)$, $0 < x < \frac{1}{4}$

(b) $f'(x) > f(x)$, $0 < x < \frac{1}{4}$ (d) $f'(x) < f(x)$, $\frac{3}{4} < x < 1$.

C: Consider the differential equation $e^x(ydx - dy) = e^{-x}(ydx + dy)$. Let $y = f(x)$ be a particular solution to this differential equation which passes through the point $(0, 2)$. Let $C: y = \log_{\frac{1}{4}}\left(x - \frac{1}{4}\right) + \frac{1}{2}\log_4(16x^2 - 8x + 1)$, be another curve

Q5. The range of the function $g(x) = \log_2(f(x))$ is

(a) $[1, \infty)$ (b) $[2, \infty)$ (c) $[0, \infty)$ (d) NOT

Q6. The area bounded by the curve C, $y = f(x)$ and the line $x = \frac{1}{4}$ & $x = 1$ is

(a) $[\sin 1 - \sin\frac{1}{4}]$ (b) $2[\sin 1 - \sin\frac{1}{4}]$

(c) $2[\sin 1 - \sin\frac{1}{4}] - \frac{3}{4}$ (d) NOT

Q7. If the area bounded by the curve $y = f(x)$, curve C, ordinate $x = \frac{1}{4}$ and the ordinate $x = a$ is $4 - \ln 4 + \frac{1}{e^{1/4}} - e^{1/4}$, then value of a is

(a) $\ln 6$ (b) $\ln 4$ (c) 4 (d) $\ln 12$.

Q8. Area bounded by the curve $y = \sqrt{\sin[x] + [\sin x]}$ (where $[\cdot]$ is GIF), lines $x = 1$ and $x = \frac{\pi}{2}$ and the x-axis is given by $\sqrt{\sin k}\left(\frac{\pi}{2} - k\right)$, then evaluate k.

Q9. If the area of the finite portion of the figure bounded by the curves $y = 2x^2e^x$ & $y = -x^3e^x$ is given by $ae^{-2} - b$, then evaluate $a - b$.

Q10. The tangent to the graph of the function $y = \sqrt[3]{x^2}$ is such that the abscissa x_0 of the point of tangency belongs to the interval $[\frac{1}{2}, 1]$. For what value of x_0 is the area $S(x_0)$ of the triangle bounded by the tangent, the x-axis, and the straight line $x = 2$ the least and what is it equal to?

Q11. Consider the two curves $C_1 : y = 1 + \cos x$ and $C_2 : y = 1 + \cos(x - \alpha)$ for $\alpha \in (0, \frac{\pi}{2}]$, $x \in [0, \pi]$. Find the value of α, for which the area of the figure bounded by the curves C_1, C_2 and $x = 0$ is same as that of the figure bounded by C_2, $y = 1$ and $x = \pi$. For this value of α, find the ratio in which the line $y = 1$ divides the area of the figure by the curves C_1, C_2 and $x = \pi$.

Q12. Find the ratio in which the curve $x = \left[\frac{25}{11} - \sin y - \frac{1}{2}\cos 2y\right]$, $y \in [0, \pi]$ divides the area bounded by the curves $y = x^3 - x^2$ & $y = x^2$, where [.] is GIF.

Q13. For what value of 'a' ($a \in [0, 1]$) does the area of the figure bounded by the graph of the function $y = f(x)$ and the straight line $x = 0$, $x = 1$, $y = f(a)$, have the greatest value and for which value does it have the least value if $f(x) = x^\alpha + 3x^\beta$, $\alpha, \beta \in R$ with $\alpha > 1$, $\beta > 1$?

Q14. solve $\frac{dy}{dx} = \frac{1+y^2}{xy(1+x^2)}$.

Q15 $\frac{dx}{x^2-xy+y^2} = \frac{dy}{2y^2-xy}$.

Q16. $yy_1 + x = \frac{1}{2}\left(\frac{x^2+y^2}{x}\right)^2$

Q17. $\left(2xy + x^2y + \frac{y^3}{3}\right)dx + (x^2+y^2)\,dy = c$.

Q18. $y_1 = \frac{(1+y)^2}{x(y+1) - x^2}$

Q19. $\frac{dy}{dx} = \frac{e^x}{2} \cdot \frac{e^{y^2/x}}{y} - \frac{1}{2}\left(\frac{x}{y} + \frac{y}{x}\right)$

Q20. $\frac{y}{x} + \frac{dy}{dx} = \frac{1}{\sin^4(xy) + \cos^2(xy)}$.

Q21. $\left(\frac{1}{y}\sin\frac{x}{y} - \frac{y}{x^2}\cos\frac{y}{x} + 1\right)dx +$

$\left(\frac{1}{x}\cos\frac{y}{x} - \frac{x}{y^2}\sin\frac{y}{x} + \frac{1}{y^2}\right)dy = 0$

Q22. $\frac{dy}{dx} = \frac{2y^2\cos x + y\sin 2x + 2\cos x \sin^2 x}{\sin^2 x}$

Q23. solve $2xyy_1 + (1+x)y^2 = e^x$ on $(0,\alpha)$ with

(A) $y = \sqrt{e}$ when $x = 1$ (B) $y = -\sqrt{e}$ when $x = 1$

(C) A finite limit as $x \to 0$.

Q24. Find all solutions of $x(x+1)y_1 + y = x(x+1)^2 e^{x^2}$ on the interval $(-1, 0)$. Prove that all solutions approach 0 as $x \to -1$, but that only one of them has a finite limit as $x \to 0$.

Q25. solve: i) $x^2 y''' = (y'')^2$

ii) $y''[1+(y')^2] = 3y'(y'')^2$

Q26. If $f: R \to [0,\infty]$ be a function satisfying $f(x+y) - f(x-y) = f(x)[f(y) - f(-y)]$, $f'(0) = \log a$, $f(0) = 1$ (for all values of a except 1), then

Solve the differential equation

$$\frac{dy}{dx} = \frac{2\{\log_a(f(x)f(y))\}^2}{(\log_a f(x) + 2)(\log_a f(y) - 2)}$$

Q27. A differentiable function f satisfies the relation $[1 + f(x) + f(y)]f(x+y) = f(x) + f(y)$ $\forall\ x, y \in R$. If $f(0) = 0$ and $f'(0) = 1$, find $f(x)$.

Q28. Find all functions f such that f' is continuous and $[f(x)]^2 = 100 + \int_0^x 2\{[f(t)]^2 + [f'(t)]^2\}dt$ $\forall\ x \in R$.

Q29. If $f: R - \{-1\} \to R$ be a differentiable function which satisfies: $f(f(x) + f(y) + x f(y)) = y + f(x) + yf(x)$ $\forall\ x, y \in R - [-1]$. Find $f(x)$?

Q30. Let $S(x) = \frac{\sin x}{x}$ if $x \neq 0$, and let $S(0) = 1$. Define $T(x) = \int_0^x S(t)dt$. Prove that the function $f(x) = x\,T(x)$ satisfies the differential equation $xy' - y = x\sin x$ on the interval $(-\infty, \infty)$ and find all solutions on this interval. Prove that the differential equation has no solution satisfying the initial condition $f(0) = 1$.

Q31. Show that the complete solution of $\frac{d^2S}{dt^2} = \frac{1}{\sqrt{S}}$ can be put in the form

$$3t = 2(\sqrt{S} - 2c_1)\sqrt{\sqrt{S} + c_1} + c_2.$$

Q32. Prove that there is exactly one function f, continuous on the positive real axis, such that $f(x) = 1 + \frac{1}{x}\int_1^x f(t)dt$, $x > 0$ and find this function.

Q33. Given a function which satisfies the differential equation $x f''(x) + 3x[f'(x)]^2 = 1 - e^x$ for all real x.

(A) If f has an extremum at a point $c \neq 0$, show that this extremum is a minimum.

(B) If f has an extremum at 0, is it a maximum or minimum? Justify your conclusion

(C) If $f(0) = f'(0) = 0$, find the smallest constant A such that $f(x) \leq Ax^2$ for all $x \geq 0$.

Q34. A dog sees a rabbit running in a straight line across an open field and gives chase. In a rectangular co-ordinate system assume

i) The rabbit is at the origin and the dog is at the point $(1, 0)$ at the instant the dog first sees the rabbit.

ii) The rabbit runs up the y-axis and the dog always runs straight for the rabbit

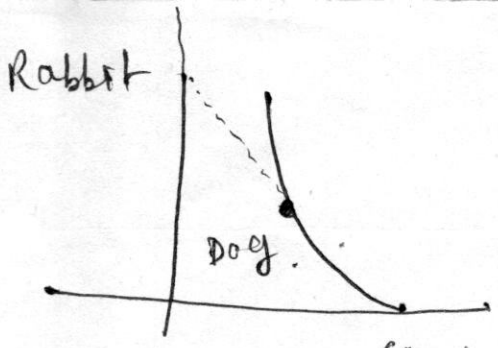
iii) The dog runs at the same speed as the rabbit

(A) Show that the dog's path is the graph of the function $y = f(x)$, where y satisfies the differential equation $x\frac{d^2y}{dx^2} = \sqrt{1+\left(\frac{dy}{dx}\right)^2}$.

(B) Determine the solution of the equation in part

(C) That satisfies the initial conditions $y = y' = 0$ when $x = 2$.

(D) Does the dog ever catch the rabbit?



Q35. Let A_r be the area of the region bounded between the curves $y^2 = (e^{-kx})x$ where $k>0$, $r \in N$ and the line $y = mx$ (where $m \neq 0$), k & m are some constants. Then which of the following is/are correct?

(A) $A_1, A_2, \ldots$ are in G.P with common ratio equals e^{2k}.

(B) $A_1, A_2, \ldots$ are in G.P with common ratio equals e^{-k}.

(C) If $\lim_{n\to\infty} \sum_{i=1}^{n} A_i = \frac{1}{48(e^{2k}-1)}$ then m equals 2

(D) If $\lim_{n\to\infty} \sum_{i=1}^{n} A_i = \frac{1}{48(e^{2k}-1)}$ then m equals 3.

Q36. Let A_r be the area of the region bounded by the curve $(6p^3 r y^2 - x)(6e^2 y - x) = 0$. Then

(A) $\sqrt{A_1}, \sqrt{A_2}, \sqrt{A_3}, \ldots$ are in G.P

(B) $\sqrt{A_1}, \sqrt{A_2}, \sqrt{A_3}, \ldots$ are in H.P

(C) $\lim_{n\to\infty} [\sqrt{A_1}\sqrt{A_2}\sqrt{A_3} + \sqrt{A_2}\sqrt{A_3}\sqrt{A_4} + \ldots n \text{ terms}] = \frac{1}{4}\left(\frac{e}{\pi}\right)^9$

(D) $\lim_{n\to\infty} [e^{-\frac{1}{\sqrt{A_1}}} + e^{-\frac{1}{\sqrt{A_2}}} + \ldots n \text{ terms}] = \left(e^{\frac{\pi^3}{e^3}} - 1\right)^{-1}$

